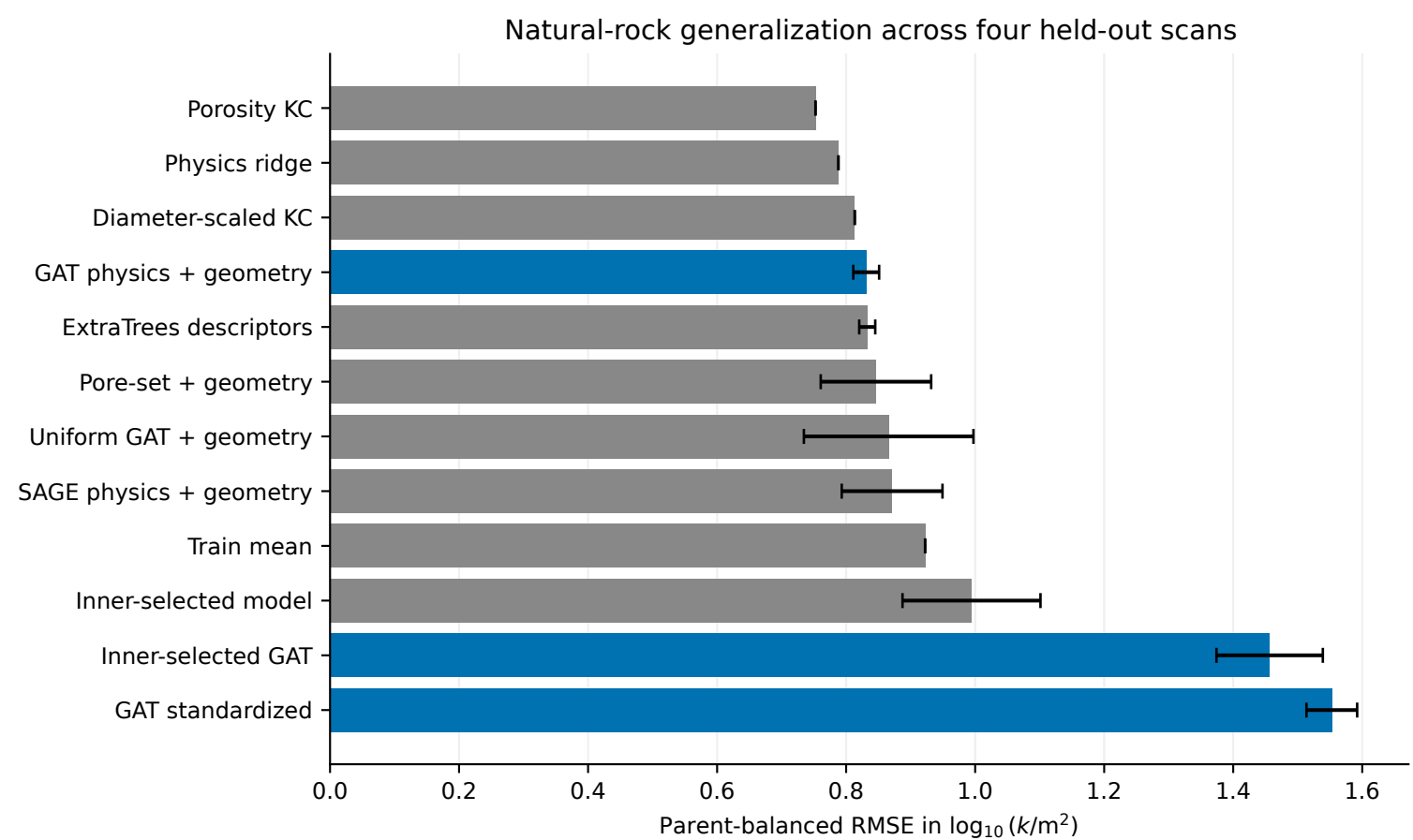
Natural-rock generalization across four held-out scans
Porosity KC
Physics ridge
Diameter-scaled KC
GAT physics + geometry
ExtraTrees descriptors
Pore-set + geometry
Uniform GAT + geometry
SAGE physics + geometry
Train mean
Inner-selected model
Inner-selected GAT
GAT standardized
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
Parent-balanced RMSE in log10 (k/m2)
Bars: mean over five seeds; whiskers: seed SD. Four natural parents, 24 crops.